A CLOSED SUBSET WITH CONNECTED INTERSECTIONS WITH CLOSURES SPLITS OFF A DECOMPOSITION OF A SPECTRAL SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the construction of the specialising sets and their constructible closedness, the partition of X from the connectedness hypothesis, and the specialisation argument giving closedness in X.

Editorial note. The label lemma-connected-topological names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Spectral spaces and the constructible topology. Topology, the definition of a spectral space and the lemma making the set of points specialising to a quasi-compact subset closed in the constructible topology. Supplies the closedness of the sets W_i in the constructible topology.
- Specialisation in the constructible topology. Topology, the lemma allowing a specialisation into a subset closed in the constructible topology to be realised by a point of that subset. Supplies the point x_1 in W_1 specialising to x, which is what makes W_1 closed in the original topology.

Source reference. Original definition 08YG: sources/stacks-chow/tags.txt, source lines 11339–11339. Prerequisite pointer invoked at proof line(s) 15077; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0903: sources/stacks-chow/tags.txt, source lines 11396–11396. Prerequisite pointer invoked at proof line(s) 15092; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A31: sources/stacks-chow/tags.txt, source lines 12724–12724. Prerequisite pointer invoked at proof line(s) 15089; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Z \subset X$ be a closed subset of a topological space X. Assume*

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(1) *X is a spectral space (Topology, Definition 08YG), and*
(2) *for $x \in X$ the intersection $Z \cap \overline{\{x\}}$ is connected (in particular nonempty).*

If $Z = Z_1 \amalg Z_2$ with Z_i closed in Z, then there exists a decomposition $X = X_1 \amalg X_2$ with X_i closed in X and $Z_i = Z \cap X_i$.

Proof. Observe that Z_i is quasi-compact. Hence the set of points W_i specializing to Z_i is closed in the constructible topology by Topology, Lemma 0A31. Assumption (2) implies that $X = W_1 \amalg W_2$. Let $x \in \overline{W_1}$. By Topology, Lemma 0903 part (1) there exists a specialization $x_1 \rightsquigarrow x$ with $x_1 \in W_1$. Thus $\overline{\{x\}} \subset \overline{\{x_1\}}$ and we see that $x \in W_1$. In other words, setting $X_i = W_i$ does the job. □